

EDUCATIONPLUS

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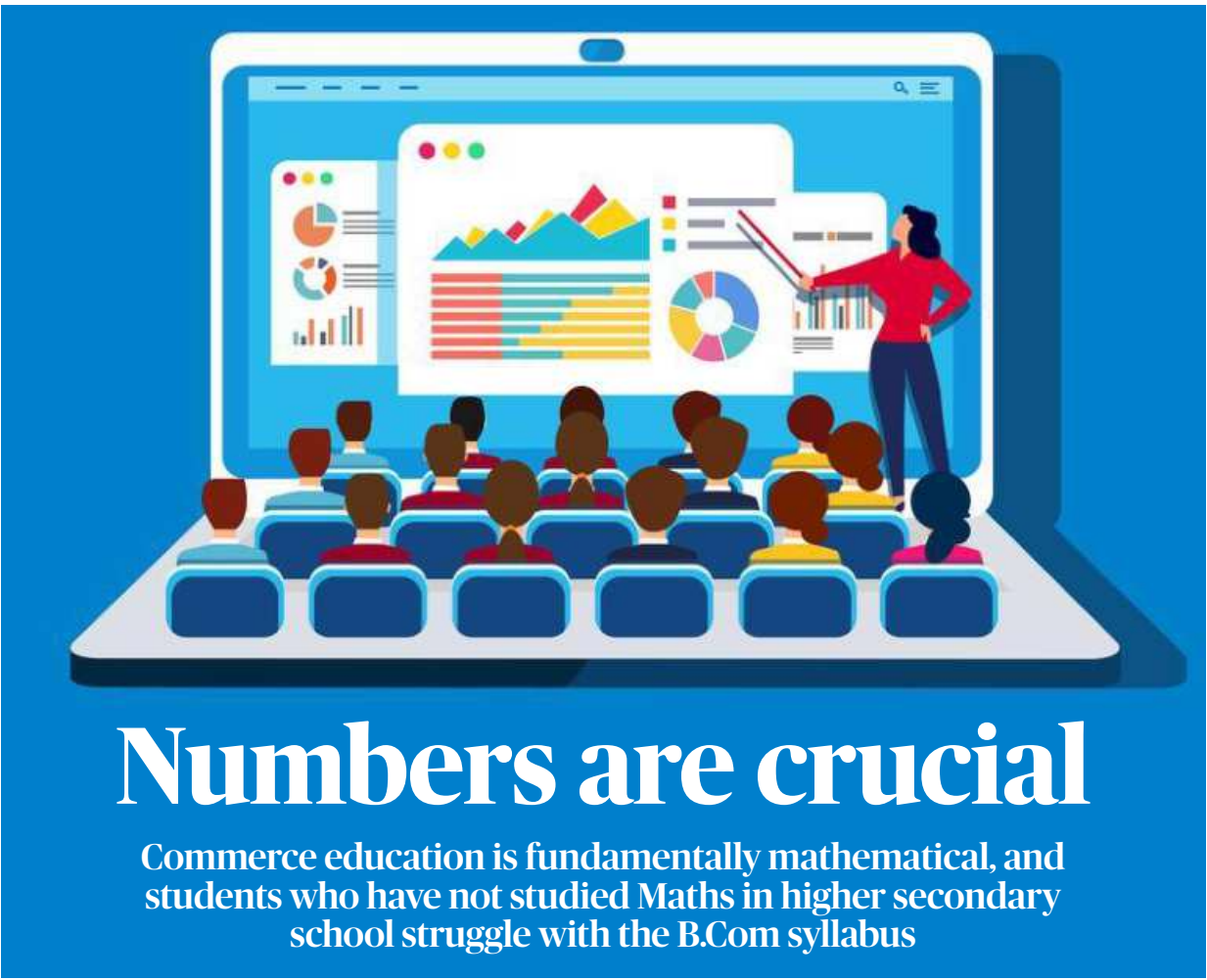
Vipin Benny

India is currently witnessing a paradox in Commerce education. Every year, lakhs of students aspiring to build careers in accounting, banking, finance, taxation, business analytics, and management enrol in B.Com programmes across thousands of colleges affiliated to central and state universities. But a growing number have never studied Maths in Classes 11 and 12 due to the National Education Policy (NEP) 2020 and its multidisciplinary spirit.

Policy in practice

What sounds like an inclusive policy on paper is, in practice, producing graduates who are structurally unprepared for the quantitative demands of modern commerce. The consequences of this are being felt in classrooms, in placement cells, and in the corridors of India's banking and financial services industry. The time has come for educators, policymakers, and institutions to confront this inconvenient reality with urgency and honesty.

Commerce education is fundamentally mathematical. The B. Com curriculum across the country includes subjects demanding rigorous numerical reasoning: Financial Accounting, Cost Accounting, Business Statistics, Economics, Corporate Finance, Income Tax, Auditing, and Business Analytics. These subjects presuppose comfort



with algebra, percentages, ratios, logarithms, and basic calculus. Academic surveys and faculty observations consistently show that students who have not studied Maths in Class 11-12 struggle with ratio analysis, index numbers, regression, and fundamental concepts like elasticity, marginal utility, and national income aggregates.

The UGC's B.Com framework under NEP 2020 intensifies this crisis by mandating Data Analytics,

Computational Finance, and Quantitative Methods as core credits. Faculty face an impossible choice: dilute the curriculum or watch unprepared students fail. Neither outcome serves the nation's educational or economic interests.

New demands

Due to India's economic transformation, the B. Com degree no longer leads to clerical roles. Today's BFSI sector, reshaped

by algorithmic trading, digital lending, GST software, and AI-driven risk assessment, demands far more. Companies expect even entry-level professionals to demonstrate Excel modelling, financial ratios, statistical forecasting, and Python proficiency.

A 2024 NSDC report revealed that 38% of Commerce and Management graduates lack quantitative skills for immediate industry deployment; strongly correlated with the rise of

non-Maths Commerce tracks at the Plus Two level. The pass rates of the CA Foundation level in Quantitative Aptitude have declined notably among non-Maths candidates, a trend ICAI has formally acknowledged.

Addressing this crisis demands a frank acknowledgement that the current approach to Maths in Commerce education is fundamentally unsustainable, and making Commerce Maths – covering Financial

Mathematics, Business Statistics, Matrices, Linear Programming, and Probability – a compulsory Higher Secondary subject for all students intending to pursue B. Com or allied programmes. Several state boards already offer or mandate Commerce Maths as a distinct paper, and early data from these states consistently indicate better performance and retention in undergraduate courses.

Simultaneously, B. Com admission norms require urgent revision. Universities should introduce a minimum Maths competency as a formal eligibility criterion or at least mandate compulsory bridge or remedial modules for students without a Maths background in the first semester. The UGC, which has been admirably proactive in reforming the curriculum under NEP 2020, must take cognisance of this structural gap and issue clear and enforceable advisories.

India's ambition to become a knowledge economy and global services hub ultimately depends on the quality of its commerce and finance workforce. That workforce is built in classrooms, and that is where Maths must reclaim its essential and indispensable place in Commerce education.

Views expressed in this article are personal

The writer is an Assistant Professor and a Research Supervisor at St. Thomas College (Autonomous), Thrissur, Kerala.

SCHOLARSHIPS

Tata Capital Pankh Scholarship

An initiative of Tata Capital
Eligibility: Indian students in Classes 11 and 12 or pursuing UG, Diploma, or ITI courses at recognised institutions, with minimum 60% marks in the previous year and annual parental income not exceeding ₹400,00.
Rewards: Up to ₹15,000 (school) and up to ₹18,000 (UG, diploma, and ITI)
Application: Online
Deadline: October 26
www.b4s.in/edge/TCP37

NSP Central Sector Scholarship of Top Class Education for SC Students

Offered by the Ministry of Social Justice and Empowerment, Government of India.
Eligibility: Students from Scheduled Caste communities who have

passed the Class 12 exam with minimum 60% marks, have secured admission in a recognised institution for a full-time UG or PG degree and have an annual family income of less than ₹800,000.
Rewards: ₹86,000 (first year) and ₹41,000 (every subsequent year of study)
Application: Online
Deadline: October 31
www.b4s.in/edge/SJEE4

NSP-ICAR Postgraduate National Talent Scholarship

Offered by the Indian Council of Agricultural Research (ICAR).
Eligibility: Students who have secured admission to approved courses in an agricultural university outside their state of domicile.
Rewards: ₹5,000 a month.
Application: Online
Deadline: October 31
www.b4s.in/edge/NTSPGI

Courtesy: Buddy4study.com

Money matters

Bitget has partnered with UNICEF to create an online financial literacy module for the Game Changers Coalition curriculum. Developed as an interactive self-paced online course, the module uses short videos, visual content, and quizzes to cover supply and demand, budgeting, credit,

loans, interest rates, and blockchain basics. The programme is designed primarily for young learners and is accessible free of charge through Agora, the UNICEF's learning platform.

Participants who complete the course will receive a certificate.

For more details about the course, visit https://tinyurl.com/476yhafc

Your future, your choice

Uncertain about your career options? Low on self-confidence? This column may help



OFF THE EDGE
Nandini Raman

I scored 98.2% in Class 12 (Humanities, CBSE) but took a gap year. I opted for the Humanities in Class 12 and am pursuing UG in Political Science. But I am exhausted by too much theory. I am preparing for CAT. Should I switch to Economics? What other alternatives do I have? Yukti

Dear Yukti,
How comfortable are you with Maths? If you did not study Maths in Class 12, or dislike Calculus and Statistics, Economics may feel more difficult than Political Science. However, if you enjoy Maths, Economics aligns well with the Quantitative Aptitude (QA) and Data Interpretation (DI) sections of CAT.
Explore a Double Major/Minor shift with a less text-heavy discipline or minors/electives such as Economics, Psychology, or Statistics. But this will depend on your university's rules. Under NEP 2020 rules, you can exit after one year with a certificate and with a diploma after two years. Your accumulated credits may help you pursue a more practical degree elsewhere. A BBA aligns strongly with your CAT goal.

But, if you can maintain a graduation score above 75–80%, I recommend finishing your B.A. while focusing on CAT preparation. Choose the route that protects your grades and leaves time for serious CAT preparation.

I am doing M.Sc. Data Science and Analytics from distance learning and also programmes on Coursera. I am also working in a research hospital and cannot do a regular degree. How can I get placed in a big tech company? Deepak

Dear Deepak,
Is the university you are doing your M.Sc. from approved by the UGC-DEB? Don't fall for the myth that hires happen only through campus placements. Major technology companies hire people through off-campus opportunities every year. Just ensure that your distance-learning M.Sc. is from a university approved by the UGC-DEB. Government regulations recognise approved distance/online degrees as equivalent to regular degrees for employment and higher education. You are already working in a research hospital so combining your practical healthcare experience with an M.Sc. and Coursera specialisations can make you a competitive candidate.

Your resume should clearly highlight relevant hospital experience and present your degree and Coursera certifications as evidence of continuous upskilling. Focus on off-campus hiring channels such as TCS iON NQT, Infosys and HCL to secure a strong baseline technology role. Also monitor entry-level opportunities at Google, Microsoft, Amazon and Meta. Use platforms such as Instahyre, Wellfound and LinkedIn Jobs. For big-tech interviews, build a strong foundation in SQL and Python/R, Data Structures and Algorithms (DSA), and

statistical thinking.

My son is in Class 8 and I am confused about the subjects he needs to take in Class 11. I want him to consider a domain, which is different and in demand so that his career can flourish. I am an IT professional and my sector is facing job losses. I don't want this to happen to him. Bhawya

Dear Bhawya,
Your child still has plenty of time before deciding on subjects for Class 11. By the time he graduates, the most resilient careers are likely to combine domain expertise with human-centric skills that automation cannot easily replace.
Promising areas for PCM/PCB in Class 11 include Climate Economy and Sustainability Engineering or Computational Biology and Bioinformatics. Economics would be an added advantage. For Humanities or Commerce background, Behavioural Economics, UX Research, Techno-Legal Compliance, and Cyber Law (B.A. LLB. or BBA.LLB.). For now, encourage him to read widely, develop critical thinking and strengthen his mathematical and communication skills. His interests and aptitudes will become clearer over time.

I am a first-year MBBS student. What skills should I build for healthcare entrepreneurship or management? Is an MBA the best route, or are there better alternatives?

Dear Abhinav,
Your interest in healthcare entrepreneurship and management from the first year gives you a significant head start.

Healthcare increasingly needs clinicians who understand business, technology and data. Focus on becoming a good doctor, while also developing non-clinical skills. Explore Digital Health and HealthTech Infrastructure, including AI-driven diagnostics, tele-health, digital therapeutics, wearable medical devices, health informatics and clinical data architecture.

Study patient triage, identify bottlenecks and examine the cost-efficiency of government versus private procurement. Medical Innovation and Bio-Design can teach you to identify clinical needs, develop solutions and navigate regulations. Build networks beyond medicine by connecting with peers in engineering, design and data science.

An MBA from a top-tier business school is a strong route into corporate management. If you aim to lead healthcare verticals, drive venture-capital investments in HealthTech or work in healthcare consulting, a general MBA can be particularly valuable. Other options include immersive Bio-Design Fellowship programmes in healthcare entrepreneurship at CFHE, IIT Hyderabad or IIT Delhi, and Master's programmes in Healthcare Administration (MHA) or Public Health (MPH) at institutions such as TISS-Mumbai, IHMR or universities abroad. Explore institute-specific need/merit scholarships, corporate scholarships and government scheme.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge

Need for timely dialogue

It is time to move beyond disciplinary boundaries and engage with fundamental questions about learning in the age of AI

S.A. Thameemul Ansari
T.A. Mohamed Mahir

The rapid emergence of Artificial Intelligence in education has compelled institutions to reconsider not only their pedagogical practices but also the philosophical foundations of learning itself. While AI promises efficiency, personalisation, and data-driven insights, its integration into educational systems is far from straightforward.

One of the most pressing challenges is not technological but institutional: departments often function in isolation, guided by distinct epistemologies and methodological commitments. In such a fragmented environment, the integration of AI risks becoming uneven, inconsistent, and, at times, counterproductive. Therefore, what is urgently required is a strategic, inclusive, and critical conversation among all stakeholders to collectively determine how AI should be integrated and to what extent.

Operate in silos

At present, educational institutions exhibit a clear divide in their approach to knowledge and data. Disciplines in Science and Technology tend to embrace data-driven decision-making, viewing data as an objective and reliable foundation for knowledge production. In contrast, fields within the Humanities and Social Sciences often approach data with scepticism, emphasising its constructed nature and the potential for manipulation. While this divergence reflects the richness of academic diversity, it becomes a barrier when operating in silos.



The lack of communication also leads to a dissipation of institutional energy. Each unit develops its own understanding of AI, adopts tools independently, and implements practices without considering the broader academic ecosystem. As a result, students receive fragmented experiences, and the institution fails to harness the full potential of AI.

To address this challenge, it is essential to bring all stakeholders – faculty, administrators, students, technologists, and policymakers – under a single umbrella for frank conversations that move beyond disciplinary boundaries and engage with fundamental questions: What do we mean by learning in the age of AI? How do we balance efficiency with depth? What are the ethical implications of delegating cognitive tasks to machines?

Perspective shift

The central focus should be a shift in perspective: from the “cost of time” to the “cost of error.” Much of the current enthusiasm for AI is driven by its ability to save time. Automated grading, content generation, and data analysis promise to re-

duce workload and increase productivity. But when AI systems produce errors –whether factual inaccuracies, biased outputs, or misleading interpretations – the consequences do not merely affect outcomes; they shape minds, influence thinking patterns, and alter the nature of human relationships.

The “cost of error” is particularly significant because it is often invisible. Unlike time, which can be measured and quantified, errors in understanding, judgment, and critical thinking are subtle and cumulative. If students become overly reliant on AI-generated responses, their engagement with knowledge may become passive, mediated by algorithms rather than driven by curiosity. Over time, this could lead to a decline in intellectual autonomy and a weakening of the very skills that education seeks to cultivate.

Moreover, the integration of AI has implications for human relationships within educational spaces. Teaching is not merely the transmission of information; it is an interaction that involves empathy, mentorship, and dialogue. If AI tools begin to replace or

mediate these interactions, students may find themselves interacting more with systems than with teachers, and the sense of community that underpins learning could be eroded.

Concrete outcomes

A strategic conversation must, therefore, prioritise critical reflection over technological enthusiasm. It should create space for dissenting voices, particularly from disciplines that question the authority of data and algorithms. At the same time, proponents of data-driven approaches can contribute valuable insights into the practical applications and efficiencies that AI offers. The goal is not to arrive at a single, uniform approach but to develop a shared understanding that respects disciplinary differences while aligning institutional goals.

Such conversations should also lead to concrete outcomes. Institutions need to develop guidelines that define the scope and limits of AI use in teaching, learning, and assessment and address issues such as academic integrity, transparency, accountability, and inclusivity. Importantly, students must be active participants in this dialogue. As primary stakeholders, their experiences, concerns, and aspirations should inform institutional decisions. This inclusion can also foster a sense of responsibility and agency, encouraging them to use technology thoughtfully rather than passively.

The urgency of this conversation cannot be overstated. As AI continues to evolve, the decisions made today will shape the future of education and, by extension, society.

The views expressed are personal

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Sampada Shevde

Picture a student sitting in a university lecture hall. She arrived early, found a seat, and opened her notebook but could not hear a word. She cannot see what is on the board. When the weight of the semester becomes too much, she has nowhere to turn. Not the helpline on the noticeboard, which she cannot read. Not the counsellor’s office, which she cannot navigate alone. She does not lack the desire to reach out. She lacks a system built to receive her.

She is a student with deafblindness, a distinct condition involving the combined loss of both hearing and vision. Her experience on India’s university campuses is far more common than the conversations around student mental health suggest.

An assumption

This is a significant moment for student wellbeing in Indian higher education. The University Grants Commission has mandated

Beyond the ramp

All support mechanisms being built on Indian campuses are for students who can hear, speak, read, and communicate conventionally. What about those who cannot?



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wellness centres, trained counsellors, and Equal Opportunity Centres at every institution.

All of it is meaningful but is built on one quiet assumption: that students in distress can ask for help in a way the system knows how to receive.

Every support mechanism now being built on Indian campuses – from helplines to counselling sessions to peer networks – is for a student who can hear, speak, read, and communicate conventionally. For most students, this assumption is invisible because it is never tested. But, for a student with deafblindness, it is the reason the door to support never opens.

India is home to an estimated 5,00,000 people living with deafblindness, and a growing number are reaching higher education. Research conducted in India identifies five dimensions through which their psychological well-being must be understood: emotional regulation, social functioning, behavioural patterns, cognitive load,

and physical well-being. Most campuses define accessibility as physical access: ramps, lifts, large print. These are necessary, but mental health accessibility asks a harder question: can the student use what is inside the building? A counselling room only works if communication is possible within it. A wellness centre only functions as a safety net if it can catch the student who falls.

Replicable model

The solutions are neither theoretical nor distant. In the U.K., universities supporting deafblind students work with trained professionals who bridge the sensory gap by conveying spoken content and social information through tactile communication. This is not a scribe or a sign language interpreter. It is a distinct role that makes campus participation possible for a student who would otherwise be entirely isolated.

India does not yet formally recognise this role, but the model is directly

replicable. The UGC and individual universities have an immediate, low-cost opportunity here: a deafblind-inclusion clause within Equal Opportunity Centre guidelines, requiring at least one trained point of contact per campus.

What is missing is not expertise or intention. It is the discipline of asking, before any support system is finalised: Does this reach every student?

If you are a student, ask your institution who its mental health systems were designed for. If you are an educator, the investment you are making in student well-being is real and it matters. The next step is making sure no student is standing outside it, unable to reach it, concluding from the silence that no help exists. The quietest students on our campuses are not the ones with nothing to say. They are the ones we have not yet learned to listen to.

The writer is Director – Education and Research, Sense International India.

The rulebook and the reason

Why India’s universities need both a good ordinance and a good heart



THE KNOWLEDGE GAP
Ram Kumar Kakani

Anyone raised on folktales would know the tale of Tenali Rama and the cat. Vijayanagara is overrun by rats, and King Krishnadevaraya decrees a cat in every household to hunt them and a cow to provide milk for the cats. The order is obeyed to the letter and it fails completely. Fattened on free milk, the cats lose all interest in rats. It takes Tenali’s mischief – a bowl of scalding milk – to send one cat back to hunting and remind the court that the point was never the milk. It was the rats.

I have been thinking about that cat while reading two very different documents this month. One is a gazette notification titled the University Grants Commission’s (Minimum Standards of Instruction for the Grant of UG and PG Degrees) Regulations, 2025. The other is an essay on why some teachers treat the classroom as a calling, not a job. One tells universities what they must do. The other reminds educators why the work matters. The perennial danger in Indian higher education is the king’s: we make the rule, then never check whether the rats are gone. Together, these documents are the antidote: one supplies the rule and the other offers its reason.

The rule

The 2025 Regulations are subtle. On assessment, they mandate that “there shall be continuous evaluation in each course in addition to semester/

year-end examination”, making continuous internal evaluation (CIE) non-negotiable. But the UGC refuses to dictate the recipe: no fixed split, no prescribed weighting. It hands that decision to “the appropriate academic body of the university” and insists only that students be told the scheme at the start of the semester, not cruelly in the week preceding examinations.

Then, two guarantees that students will recognise as fairness. Marking stays confidential but the system must be “sufficiently transparent”, and a student “shall be given access to his/her answer script”. The heart of it: institutions “shall accord priority to continuous formative assessment”. The message: Delhi sets the guardrails; the driving is left to you.

This is autonomy with teeth. The UGC may debar a defaulting institution but autonomy, as every Board of Studies knows, is a double-edged gift. Give a committee freedom to design its own CIE and you might get imaginative, outcome-linked assessment or four hurriedly scribbled “assignment marks” from a teacher who would rather be elsewhere. The regulation can compel the structure of fairness. It cannot legislate the spirit of it. That is precisely where the second document walks in.

The reason

The essay on teaching as vocation sounds sentimental until you notice how practical it is. A profession can be “chosen, practised competently, and set aside at day’s end”. A vocation stays with you, shaping how you invest in the people

around you. The teacher who treats the work as a calling does not need a circular to give honest feedback; she gives it because a half-read script betrays a student who trusted her with a semester.

A regulation can hand out the cats and the cows. Only someone who remembers the rats can make the hunting happen.

For students, the most useful sentence in the 2025 Regulations is this: you are entitled to see your own evaluated answer script and to know your assessment scheme before the semester begins (not after!). Ask for both. That is the floor the rulebook guarantees; the vocational teacher supplies the ceiling with feedback that helps you improve. Rights on paper mean nothing if the marker couldn’t care less. Both must be present or the rats stay put.

So, one thing to do on Monday. If you are a student, ask to see your assessed script and what it was measured against. If you teach, redesign one assessment to measure learning, not compliance. If you lead, ask what your dashboards actually measure: the milk, or the rats.

Get the rulebook without the reason and we keep feeding contented cats while the rats multiply. Get both, and one honest answer script at a time, the knowledge gap begins to narrow. The rules are now written. Whether the hunting happens is up to the person holding the pen over a student’s script.

Views expressed are personal

The writer is the Vice-Chancellor of RV University, Karnataka



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Design the future

What does a career in Product Design entail?

A. Balasubramaniam

The Indian design environment is experiencing a massive transformation. Whether it is IT-based startups or e-commerce firms, the healthcare industry or the consumer goods sector, design skills have become crucial in making one’s product stand out in a highly competitive environment.

Product Design is the process of creating products and experiences that meet user needs while balancing functionality, aesthetics, and usability. It is a complex process involving research, idea generation, prototyping, testing, and improvements to come up with efficient designs.

Today, Product Design is morphing and evolving as a discipline. Besides physical, tangible products such as automobiles, kitchenware, furniture, and so on, it also encompasses digital platforms, mobile



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apps, websites, and software services as well. Product Design has gained rapid significance thanks to the growing digital economy. Companies are now paying attention to design details to provide enhanced customer satisfaction, improve engagement and boost their businesses’ growth.

What to study

Students interested in pursuing product design should enrol in a Bachelor’s programme in Design (B.Des.). The number of institutions that offer B.Des.

has also significantly increased over the years. This course helps students gain knowledge of the design process, product development, material technology, manufacturing process, and ergonomics. Master’s and Doctoral programmes can be useful for specialisation and to enhance knowledge and skills.

Designing products requires a combination of creative thinking, technical skills, and problem-solving capabilities. Creativity is the differentiator since designers have to come up

In the land of the maple

Tips for students who are considering Canada as a study-abroad destination



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Financial plan: Beyond tuition fees, students need to budget for accommodation, food, transportation, healthcare, and other expenses. They also need to meet proof-of-funds requirements as part of the study permit process. Canada has a range of scholarships and funding opportunities, alongside institution-specific support.

For example, the University of Toronto up to \$25 million in scholarships. Students should explore a range of study destinations across Canada, as living expenses vary by region.

Visa pathways and post-study options: The first step is to secure ad-

mission in a Designated Learning Institution and receive a Letter of Acceptance before applying for a study permit.

Common requirements include proof of funds, a valid passport, and academic records. In some cases, biometrics, a medical exam or a Provincial Attestation Letter may also be necessary.

Canada also offers post-study work opportunities through the Post-Graduation Work Permit (PGWP), allowing eligible graduates from approved DLIs to work in Canada for up to three years, depending on the programme. Recent initiatives, such as Canada’s International Talent At-

traction Strategy, have opened more opportunities in high-priority sectors such as healthcare, STEM, education, agriculture and skilled trades.

Housing: Given the high demand for housing and limited availability, the search for accommodation must begin early. Consider university residences, homestays, shared apartments and private accommodations depending on your budget and convenience. Compare costs, check commute times, and research neighbourhood safety before making a decision.

Although most institutions have September and January intakes, students should start investigating universities, scholarships, housing, and English proficiency testing needs at least 10-12 months before they enter the institution.

Mistakes to avoid: Many students make the mistake of applying to unrelated programmes to improve their admission chances. Admissions teams value clarity and consistency, so your academic background and chosen course should align naturally.

Students should also avoid delaying visa preparation or overlooking housing arrangements. Do not approach studying abroad as a rushed decision.

The writer is Country Lead, Duolingo English Test

are excellent. From toys to transportation, healthcare and home appliances, and even mobile apps and websites, opportunities are on the rise, especially with the integration of AI and digital technologies.

Serious about a career in Product Design? Check if you have the following traits:

- Are you fascinated by design? Do you spend your time making sketches of various objects?

- Do you tinker around the house and make things on your own?

- Do people ask for your opinions when choosing appliances or gadgets?

- Are you constantly showing others how to use their phones or keeping them secure?

- Are you always taking apart toys or other products to see how they work?

If you answered yes to all of these, then consider a career in Product Design.

The writer is Director, Institute of Design, JK Lakshmiipat University.

ON THE SHELF



■ Hazrat Mahal: The Rebel Begam of Awadh

The daughter of a slave (likely of African descent), a courtesan in Wajid Ali Shah’s Pari Khana before becoming one of his many wives, mother of his son, de facto leader of Awadh’s rebellion against the East India Company ... Begum Hazrat Mahal is an elusive figure in Indian history.

At the centre of her brief rule is one of the most dramatic episodes in the history of British India: the siege of the Lucknow Residency. Drawing from decades of research and previously undiscovered archival material, historian Rosie Llewellyn-Jones renders the complexities of 19th century Awadh through the story of its fiercest and most unlikely leader.

Author: Rosie Llewellyn-Jones
Publisher: Pan Macmillan
Price: ₹799